SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA1716	Course Title:	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	Duration / Marks:	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)
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Section / Batch:	2026	Date:	9.

Code of Conduct
I M. Srilekha 1919260007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: M. Srilekha

Instructions
Read each question carefully before answering.
Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	Q3
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	Q1, Q2
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	Q1, Q4

Annexure A – Analytical Problem Solving

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

Agent X (simple reflex): if the current square is Dirty, Suck; otherwise move to the other square.

Agent Y (reflex with internal state): if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given.

Answer:

Performance measure: +1 for every clean square at the end of each time step; penalty p = 1 for every Left/Right move.

Environment: Two-room vacuum world with squares A and B; both initially Dirty and dirt does not reappear.

Actuators: Suck, Right, Left, NoOp.

Sensors: Perceives its current location and whether the current square is Dirty or Clean.

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1.

Answer:

Agent X starts at A. It Sucks a dirty square; when the current square is clean, it moves to the other square.

Step 1: Suck → clean squares: A; move penalty: 0; step score: 1.

Step 2: Right → clean squares: A; move penalty: 1; step score: 0.

Step 3: Suck → clean squares: A, B; move penalty: 0; step score: 2.

Step 4: Left → clean squares: A, B; move penalty: 1; step score: 1.

Step 5: Right → clean squares: A, B; move penalty: 1; step score: 1.

Step 6: Left → clean squares: A, B; move penalty: 1; step score: 1.

Total clean-square points = 10.

Total move penalty = 4.

Total score = 10 – 4 = 6.

(c) Trace Agent Y in the same way and find its total score for p = 1.

Answer:

Agent Y remembers that both squares have been cleaned. After cleaning both squares it uses NoOp instead of moving unnecessarily.

Step 1: Suck → clean squares: A; move penalty: 0; step score: 1.

Step 2: Right → clean squares: A; move penalty: 1; step score: 0.

Step 3: Suck → clean squares: A, B; move penalty: 0; step score: 2.

Step 4: NoOp → clean squares: A, B; move penalty: 0; step score: 2.

Step 5: NoOp → clean squares: A, B; move penalty: 0; step score: 2.

Step 6: NoOp → clean squares: A, B; move penalty: 0; step score: 2.

Total clean-square points = 10.

Total move penalty = 1.

Total score = 10 – 1 = 9.10.26

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X?

Answer:

Agent X: X(p) = 10 – 4p.

Agent Y: Y(p) = 10 – p.

For Y to be strictly better:

10 – p > 10 – 4p

3p > 0

p > 0.

Therefore, Y is strictly better for every positive movement penalty p. At p = 0, both agents have the same score.

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step.

Answer:

Agent Y is rational for the given performance measure because it cleans both squares and then avoids unnecessary movements, reducing the movement penalty. The percept sequence tells the agent what it currently sees, while Y's internal state also remembers that both squares have already been cleaned. Agent X cannot match Y because it has no memory; after seeing a clean square, it always moves to the other square even when both squares are already clean.

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

E1: An autonomous taxi driving through city traffic.

E2: A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.

E3: A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each.

Answer:

E1 – Autonomous taxi:

• Observability: Partially observable – traffic and other drivers' intentions are not fully known.

• Agents: Multi-agent – other drivers and road users act independently.

• Determinism: Stochastic – traffic events are uncertain.

• Episodic/Sequential: Sequential – current decisions affect later states.

• Static/Dynamic: Dynamic – traffic changes continuously.

• Discrete/Continuous: Continuous – position, speed and time vary continuously.

E2 – Conveyor-belt part-picking robot:

• Observability: Partially observable – camera observations may be incomplete/noisy.

• Agents: Single agent – the robot is the main decision-maker.

• Determinism: Stochastic – detection and picking outcomes can be uncertain.

• Episodic/Sequential: Sequential – actions affect later decisions.

• Static/Dynamic: Dynamic – parts move continuously.

• Discrete/Continuous: Continuous – position and motion vary continuously.

E3 – Chess-playing agent:

• Observability: Fully observable – the complete board is visible.

• Agents: Multi-agent – an opponent also makes decisions.

• Determinism: Deterministic – legal moves have predictable board effects.

• Episodic/Sequential: Sequential – each move affects future positions.

• Static/Dynamic: Semi-dynamic – the board is static but the clock continues.

• Discrete/Continuous: Discrete – moves and board positions are discrete.

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest.

Answer:

E1 Taxi = 1 + 1 + 1 + 1 + 1 + 1 = 6.

E2 Robot = 1 + 0 + 1 + 1 + 1 + 1 = 5.

E3 Chess = 0 + 1 + 0 + 1 + 0.5 + 0 = 2.5.

Ranking from hardest to easiest:

E1 (6) > E2 (5) > E3 (2.5).

Therefore, E1 is the hardest and E3 is the easiest.

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification.

Answer:

Hardest – E1 (autonomous taxi): A utility-based agent is most suitable because it must choose among uncertain routes while balancing travel time, safety, traffic and other performance factors.

Easiest – E3 (chess): A goal-based agent is suitable because it has a clear goal of achieving checkmate and can evaluate possible sequences of moves toward that goal.

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be.

Answer:

Environment properties determine the complexity of the agent program. Partial observability requires maintaining information about hidden states; stochastic environments require handling uncertainty; sequential and dynamic environments require planning and continuous decision-making. Multi-agent environments also require considering other agents. Therefore, E1 needs a more complex agent than E3 because it combines the harder properties.

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system.

Answer:

Correct-identification rate = (Correctly identified / 50) × 100.

Fooling rate = 100% – Correct-identification rate.

ELIZA: 47/50 × 100 = 94% correct; fooling rate = 6%.

Chatbot B: 36/50 × 100 = 72% correct; fooling rate = 28%.

System C: 29/50 × 100 = 58% correct; fooling rate = 42%.

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level.

Answer:

The 70% criterion means the interrogator should have no more than a 70% chance of correctly identifying the machine.

ELIZA: 94% correct → does not satisfy; 44 percentage points above the 50% guessing level.

Chatbot B: 72% correct → does not satisfy; 22 percentage points above the 50% guessing level.

System C: 58% correct → satisfies; 8 percentage points above the 50% guessing level.

Therefore, only System C satisfies the 70% criterion.

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points.

Answer:

ELIZA fooling rate = 6%.

System C fooling rate = 42%.

Improvement from ELIZA to System C = 42% – 6% = 36 percentage points.

Ratio = 42/6 = 7.

Therefore, System C's fooling rate is 7 times ELIZA's.

Chatbot B fooling rate = 28%.

Improvement from Chatbot B to System C = 42% – 28% = 14 percentage points.

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument).

Answer:

The Turing Test measures whether a machine can act humanly in conversation. A high fooling rate only shows that judges often mistake the machine for a human; it does not prove that the system truly thinks, understands or acts rationally. This illustrates the distinction between acting humanly and thinking rationally. The Chinese Room argument also shows that producing convincing language behavior does not necessarily prove genuine understanding.

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$U = 100 - T - 20 \cdot L$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips.

Answer:

Utility formula: $U = 100 - T - 20L$, where L = 1 if arrival is after 55 minutes, otherwise L = 0.

R1 – Highway:

Normal: T = 30, L = 0 $\rightarrow$ U = 100 – 30 = 70.

Disrupted: T = 70, L = 1 $\rightarrow$ U = 100 – 70 – 20 = 10 (late).

R2 – City road:

Normal: T = 45, L = 0 $\rightarrow$ U = 100 – 45 = 55.

Disrupted: T = 60, L = 1 $\rightarrow$ U = 100 – 60 – 20 = 20 (late).

R3 – Ring road:

Normal: T = 50, L = 0 $\rightarrow$ U = 100 – 50 = 50.

Disrupted: no disruption; T = 50, L = 0 $\rightarrow$ U = 50.

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose.

Answer:

Expected utility = P(normal)U(normal) + P(disruption)U(disruption).

R1: EU = 0.70(70) + 0.30(10) = 49 + 3 = 52.

R2: EU = 0.90(55) + 0.10(20) = 49.5 + 2 = 51.5.

R3: EU = 1(50) = 50.

Therefore, the rational agent should choose R1 because it has the highest expected utility, 52.

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route.

Answer:

The new percept raises R1's disruption probability to 0.80, so normal probability = 0.20.

EU(R1) = 0.20(70) + 0.80(10)

= 14 + 8

= 22.

R2 remains 51.5 and R3 remains 50.

Therefore, the rational choice changes from R1 to R2.

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent.

Answer:

From part (b), the rational choice was R1. If disruption occurs, T = 70 minutes and the taxi is late.

Actual utility = 100 – 70 – 20(1) = 10.

The agent was not irrational. It chose R1 because R1 had the highest expected utility using the information available at that time. An omniscient agent would know in advance that the disruption would occur and could choose the best route for that known outcome. Rationality uses available percepts; omniscience assumes knowledge of the actual future.

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change.

Answer:

Rationality depends on the percept sequence because the agent's decision is based on the information available to it. It also depends on the performance measure because the agent tries to maximize utility. In part (c), the roads did not change, but the radio alert changed the information by increasing the probability of disruption on R1. Therefore, the expected utility changed and the rational decision changed from R1 to R2.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____